Course: DATA602 - Principles of Data Science
Assignment: Paper Writing 01
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Date: September 15, 2026

A. (Points 50): Paper Review

Paper: *Data Science and Analytics: An Overview from Data-Driven Smart Computing, Decision-Making and Applications Perspective*

Summarize the Key Points: What are the main objectives of the paper?

One of the main objectives of this paper on advanced analytical techniques of data science is to make us understand various terms which come under the data science umbrella and various kinds of data we collect and face in our daily life. Knowledge can be collected from this data and converted into insights based on the real world problem we have at hand also using machine learning and deep learning. Another important objective is to examine various methods of analytics namely descriptive, diagnostic, predictive and prescriptive giving us a clear understanding of each of them along with various machine learning models like regression, classification, clustering, deep learning, time series forecasting. Using machine learning models we can automate the computation and build intelligent systems. It emphasises that preparing and understanding data is key to how these data-driven models work and give out useful insights.

This paper serves as a guide for both researchers and students pursuing this field and would like to grow in it. Also discussed were the business problems and how their solutions can be found through data science, data analysis. Also the author discussed various beneficial outcomes of using these methods and also the challenges the author faced but that being said the challenges also give us new research opportunities which we could ponder upon and make new discoveries. As said, the research the author has done had a positive yield.

TGDS Model: What is the TGDS (Theory-Guided Data Science) model, and what is its significance in the field of data science?

TGDS was proposed by Karpatne A, Atluri G, Faghmous JH, Steinbach M, Banerjee A, Ganguly A, Shekhar S, Samatova N, Kumar V in 2017. Theory Guided Data Science integrates scientific theories and domain knowledge into data-driven machine learning to create models which are both predictive and scientifically consistent. This was done to improve model generalizability, interpretability and robustness particularly in scientific domains as data scarcity can be a problem here. By embedding theoretical constraints while designing the model or during optimization TGDS prevents the learning of invalid or questionable relations and focuses on the domain insights that are grounded by real world physics or expertise.

Impact on Future Advancements: Reflect on how the paper's focus on real-world application domains of data science could impact future advancements in the field. Provide specific examples mentioned in the paper.

Medical or health data science is one of the most notable fields in data science as research in this field can yield many outcomes which benefit us like prevent preventable disease, minimize medical cost, predict infectious outbreaks before they happen and improve the quality of life for patients. Health data science modeling can play an important role in the future, by analyzing the current

and historical data we can predict trends. In the future we may be able develop models which are interpretable and follow strict rules and regulations of healthcare and provide correct insights. Business data science is basically studying the historical business or e-commerce data to generate more insights of how to improve the business, create strategies. Data scientists can create data-driven models which can predict customer behavior, identify trends which may appear in the near future. Further in the future business automation and smart systems can be made for intelligence and decision making.

Cybersecurity is another one of the most important domains as of right now in the technical industry as defending your computer or network from malware/virus is important. Analyzing the historical data from this and analyzing has given us models that can identify threats, finding insider threats and predicting bad network traffic and keep people safe on the internet. Data Science modeling can predict the threats better than data science modeling, real time streaming and unsupervised anomaly detection as well to combat attackers.

Types of Data: Differentiate between the four types of data mentioned: structured, unstructured, semi-structured, and metadata. Provide examples of each type.

Data can be classified in 4 categories namely - Structured, Unstructured, Semi-Structured and Metadata. Structured data is a well defined data structure which has a specific structure and a predefined format. Unstructured data has no fixed format; it can come in any shape size and is not organized. Semi-Structured is like a combination of structured and unstructured data which means it's not exactly formatted but it is organized in some places where necessary. Metadata is the data about the data, information about the data which is necessary for it to be viable.

Example: Structured Data- dates, addresses, bank account information; Unstructured Data- data collected from sensors, wikipedia, videos, images, pdfs,ets; Semi-Structured Data- HTML, XML, JSON documents NoSQL databases; Metadata: file size, file type, creation date and time, author, last modification date and time,

Analytics in Decision-Making: How can organizations use a combination of descriptive, diagnostic, predictive, and prescriptive analytics to make more informed business decisions? Provide a hypothetical example where all four types of analytics are utilized.

Descriptive Analytics says "what happened in the past ?", Diagnostic Analytics says"why did it happen?", Predictive Analytics says "what will happen in the future? ", Prescriptive Analytics says "what actions should we take next".

Example: Player retention and live churn rate in a game:- It was noted after descriptive analytics that average playtime dropped by 40% and weekly login frequency dropped from 8 to 3 days once they reached level 50. After Diagnostic analytics it was noted that it happened so because it took an average of 25 hours of grinding to go up by 1 level compared to 3 - 7 hours in previous levels causing an 75% drop in mission completion rates. After Predictive Analytics it was observed that if the same pattern goes on 85% probability of permanent churning within 14 days leading to 30% loss in user base and a $600,000 loss over the next quarter. Prescriptive Analytics says that in order to reduce the churn rate and increase player retention they should increase the maximum level achievable from 60 to 100 and give a 35% boost to monster drop rate and granting bonus rewards every 5 levels, aiming to recover 22% retention.

Real-World Applications: Identify two compelling real-world applications from the "Real-World Application Domains" section. Explain what makes them intriguing and how you would leverage machine learning to tackle their challenges. Which specific models or methods would you apply and why?

Smart cities or urban data science : It is very intriguing because modern cities produce great volumes of dynamic spatio-temporal data from road sensors, gps amd surveillance systems. The challenge lies in managing the severe traffic congestion in major blocks of these cities, excess fuel emissions due to construction work blocking roads. To tackle this problem we can build an intelligent system which trains on historical data and predicst the traffic flow and considers all parameters for routing traffic signals and reducing congestion. Gradient boosting decision trees and time series forecasting models will work best traffic control, travel time forecasting, peak hours and consider weather impacts and construction blocakages. Neural Networks will work best for routing in road networks by making a graph like structure. By combining these two we can have a graph based system with traffic forecasting and adaptive signals.

Cybersecurity data science : It is the most intriguing because cybersecurity is the shield of computing and networks as the age of AI and AI developed applications lack the security measures needed for threat blocking. Cybersecurity threats are not static if one attack fails the attackers might learn from it and use another stratergy to attack the system/network, therefore there is a need to develop models by which our cybersecurity programs can learn new patterns and come up ways to tackle them. We can make a model which can automatically detect behavior anomaly across the network in real time. The use of Unsupervised anomaly detection will work great here as we dont have any previous data on new behaviours we can determine them by differentiating them from normal behaviour. Ensemble learning like random forest can help us design models on threats which are already identified to make an even robust models.

B. (Points 20): Data Ethics Case Study

Case Summary & Reference

Clearview AI built an illegal database of over 30 - 40 billion face images by scraping photos from the public internet, social media, news sites etc. Ruins the deep learning technology the company used these images to make biometric vector maps (fingerprints) to build an reverse image search engine, marketed to law enforcement anegcies and private organizations. In September 2024 the Dutch Data Protection Authorities levied a 30.5 million ($33.7 million) fine against Clearview AI, classifying the facial indexing as illegal database. It was said that the images used weren't used with their consent and that they were put in mass surveillance. THis penalty followed a $50+ million nationwide class action settlement in Illinois under Biometric Information Privacy Act.

Reference: Source: Associated Press (AP News), Author: Mike Corder, September 3, 2024, *Clearview AI fined $33.7 million by Dutch data protection watchdog over 'illegal database' of faces*,
Link: Associated Press Article Link

What ethical principle(s) were violated?

Facial images were scrapped from the internet, social media without peoples consent and no formal information was provided. The company worked under the assumption that online availablility means its free to use by anyone for anything. This violates the principle of contextual integrity. People had no way to know if their images were in the database, lack of transparency. Some people were made search targets by law enforcement agencies when they were innocent because their images were found in the database and were used to match with the actual culprit, this infringes their civl liberties.

What should have been done differently to avoid the violation?

Facial Images data wil require strict authorization because facial biometric data is immutable and cannot be treated like regular text. The company should have conducted a third party civil rights and ethical risk assesment to evaluate the risk of false positive arrests,stalker usage and unauthorized mass surveillance. The database used for this should be minimized if the use case was law enforcement related then a constrained database of past criminals should be made and used to train the model.

How can this case guide future responsible data science practices?

Facial Image is not just like text i has your biometric data and your facial fingerpint and are sensitive genetic markers. Responsible AI developers and practitioners should be aware of using documented and licensed data. Relying on this kind of data can get you i trouble as you can see.